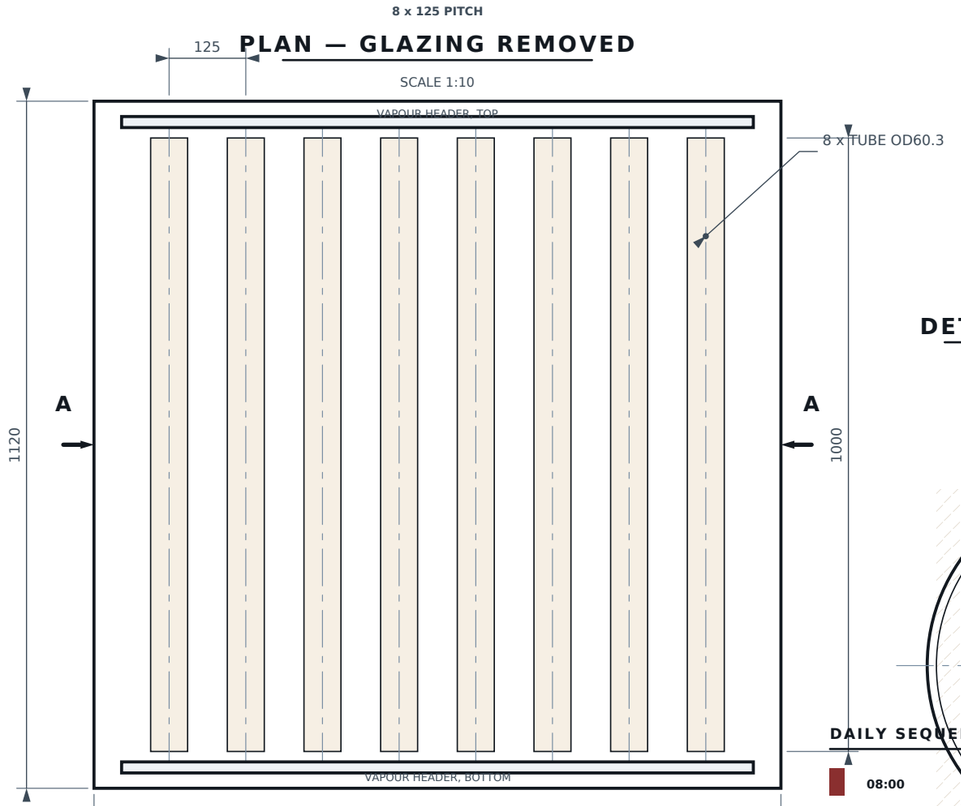
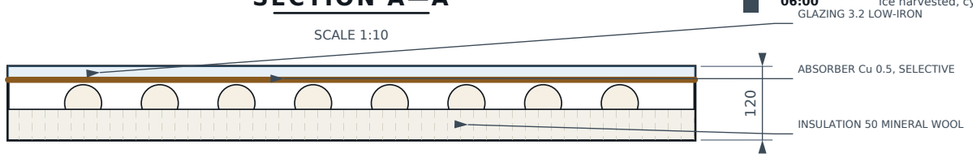
## PLAN — GLAZING REMOVED

SCALE 1:10



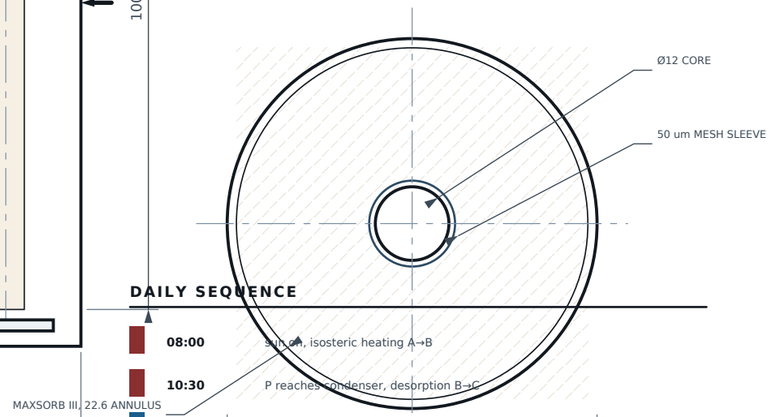

## SECTION A—A

SCALE 1:10




## DETAIL B — TUBE SECTION

SCALE 1:1




## DAILY SEQUENCE

| | | |
|---|---|---|
| | **08:00** | sun on, isosteric heating A→B |
| | **10:30** | P reaches condenser, desorption B→C |
| | **16:00** | sun off, isosteric cooling C→D |
| | **18:00** | P falls to evaporator, adsorption D→A |
| | **06:00** | ice harvested, cycle repeats |

| TUBE DATA | |
|---|---|
| carbon per tube | 0.61 kg |
| packed volume | 16.4 L |
| fill of available | 83 % |
| annulus | 22.6 mm |
| diffusion time | 11.8 min |
| operating pressure | 8.4 bar |
| **stagnation design** | **14.0 bar** |
| hoop at stagnation | 26.7 MPa |
| **margin on SS 316L** | **4.9 x** |

### NOTES

1. MATL: tubes and headers SS 316L. Absorber plate Cu, selective coating a 0.95 / e 0.10.
2. STAGNATION IS THE DESIGN CASE. A solar collector will stagnate; at 150 C methanol reaches 14.0 bar.
3. No pressure relief to atmosphere anywhere on this assembly. The tube contains stagnation instead.
4. All joints TIG welded, full penetration. No threaded or elastomer joint on the wetted side.
5. Bake carbon 200 C / 8 h under vacuum before filling; fill under dry nitrogen.
6. Core perforation 2.0 dia, 30 % open, sleeved in 50 um mesh to retain fines.

| CRITICAL CHARACTERISTICS | | | |
|---|---|---|---|
| **FEATURE** | **NOMINAL** | **TOL** | **CRITICAL TO** |
| Tube wall | 1.5 | -0 / +0.15 | stagnation containment |
| Weld, tube to header | full penetration | 100 % dye pen | hermeticity |
| Absorber-to-tube bond | continuous | ultrasonic | collector efficiency |
| Carbon mass per tube | 0.61 kg | ±2 % | cycle capacity |

**INDIA-VC THERMAL HARDWARE** — ISO 128 / ISO 2768-mK

TITLE
**ADSORBER-COLLECTOR**
SS 316L / 8 TUBES OD60.3 / 4.9 kg CARBON / DESIGN 14 bar

SCALE **1:10** — THIRD ANGLE PROJECTION
UNITS **mm**

DRAWING No. **SI-101**
REV **A**
SHEET **2 / 6**

| DRAWN | CHECKED | MATERIAL | FINISH | DATE |
|---|---|---|---|---|
| AI-ASSISTED DRAFT | — PENDING — | SEE BOM | SEE NOTES | 2026-09-15 |